Title (125 characters max)

Photogrammetry of brain dissections: automatic mesh closing works, but acquisition limits volume measurement

Impact Statement (40 words max)

Photogrammetric meshes of brain dissections can be closed automatically, but their scanned edge moves by millimetres between epochs, changing volumes as much as one epoch of dissection: acquisition must be controlled before these meshes serve as quantitative references.

Synopsis (100 words max)

Motivation: Photogrammetric meshes of brain dissections could serve as quantitative references for diffusion MRI, but they are open surfaces with scanning defects.

Goal(s): Close them into watertight volumes and measure how far these volumes can be trusted.

Approach: We repaired and closed all 130 meshes of the public BraDiPho archive, compared closing methods and checked that volume decreases along each dissection.

Results: Every mesh became watertight, and closing methods changed volumes by 0.45% on average. Yet volume increased in 15 of 104 epoch-to-epoch changes, by up to 15%, mostly because the scanned edge moved 2 to 5 mm outward.

Introduction

Photogrammetry records a brain dissection as a series of textured 3D meshes[1,2]: after each removal of tissue, the specimen is scanned again, giving one mesh per epoch. The BraDiPho archive[1] shares such meshes to compare dissected pathways with diffusion MRI tractography, and photographs of brain slices registered to surface scans already yield regional volumes[3]. Measuring the volume removed at each epoch requires watertight surfaces. However, each mesh is open on the unphotographed face and contains scanning defects (inverted faces, folds, floating debris), and a watertight mesh is not necessarily a correct one (Figure 1).

We closed the whole archive and asked: (1) can the meshes be closed automatically; (2) how much does the volume depend on how they are closed; and (3) does the volume decrease at every epoch, as removing tissue requires? We report what works and what does not.

Methods

Data. We used all 130 epochs of the 12 BraDiPho specimens[1]. Undocumented "b" variants were excluded from comparisons between epochs, leaving 104 epoch-to-epoch changes.

Repair and closing (Python, VTK; Figure 2). (1) Duplicated faces, small flaps and small fragments are removed, non-manifold edges are split, and faces are oriented outward. (2) Each opening, bounded by a loop, is closed by a cap. A plane is fitted to the loop; if the loop projects onto it without crossing itself (after straightening small folds of up to 20 vertices), a planar cap is triangulated[4] on the outer side of the opening. Otherwise, a smooth curved cap is stretched across the loop like a soap film (harmonic interpolation). (3) Remaining defects and debris are removed and parts reoriented; volume is computed only for watertight meshes.

Evaluation. Baselines: a fan closing of the same loops, two variants of fold straightening (A: always planar; B: planar only if the cap adds at most 5% of volume) and MeshFix[5], an independent repair tool. We measured (a) watertightness; (b) the closing uncertainty, i.e. the volume difference between

methods; (c) visual defects in every epoch; and (d) plausibility: dissection only removes tissue, so volume should decrease. Each increase was compared with 7 normal decreases (controls) for scale (similarity ICP), displacement of the main cap and volume added by the caps.

Results

What works. All 130 epochs became watertight (Figure 2). Straightening small folds reduced the epochs needing a curved cap from 38 to 12. Volumes differed from the fan closing by 0.45% and from MeshFix by 0.34% on average (maxima 3.4% and 4.6%; Figure 3b). After corrections, the epochs with visual defects dropped from 29 to 12.

What works poorly. Most of these 12 epochs (7 from one specimen) have truly folded openings, where the curved cap folds onto itself (Figure 4a); others have small holes just under the planar cap (Figure 4b). On four deeply hollowed openings, variants A and B differed by 5.7 to 11%: there, the volume depends on an arbitrary choice.

What does not work. The median volume removed per epoch was 8.7% (range 0.5–37.5%). Yet volume *increased* in 15 of 104 changes, by up to 15.3%, and 21 changes were smaller than the closing uncertainty (Figure 3a). Scale differences ($\leq$0.8%) did not explain them. In 9 of the 15, including the three largest, the main cap lay 2.2 to 5.1 mm further out, adding 19 to 57 cm^3, whereas it moved by -2.2 to $+1.9$ mm in the controls (Figure 5). Two others followed a poorly closed epoch, and four were within the closing uncertainty. MeshFix found exactly the same 15 increases, up to 15.4%.

Discussion

Only visual inspection revealed folds and inverted debris. The closing uncertainty (about 3%, up to 11%) cannot explain increases of 13 to 15%: the scanned edge itself moves, so any closing method inherits the shift. Each millimetre changes the volume by 2 to 4%, comparable to the tissue removed in a typical epoch. The limiting factor is the acquisition, not the closing.

With existing data such as BraDiPho, we suggest planar caps where possible, a visual check of every epoch, and reporting the closing uncertainty with each volume, flagging increases and changes within it. **For new acquisitions**, we recommend photographing both faces at every epoch, fixing the specimen on a rigid support with markers and a scale reference, and adding structured-light or lidar scanning, possibly less prone to floating fragments than stereo photogrammetry. For now, dissection photogrammetry is well suited to visualization, anatomy teaching and qualitative comparison with tractography; quantitative volumes are close, but not yet reliable.

Figure captions

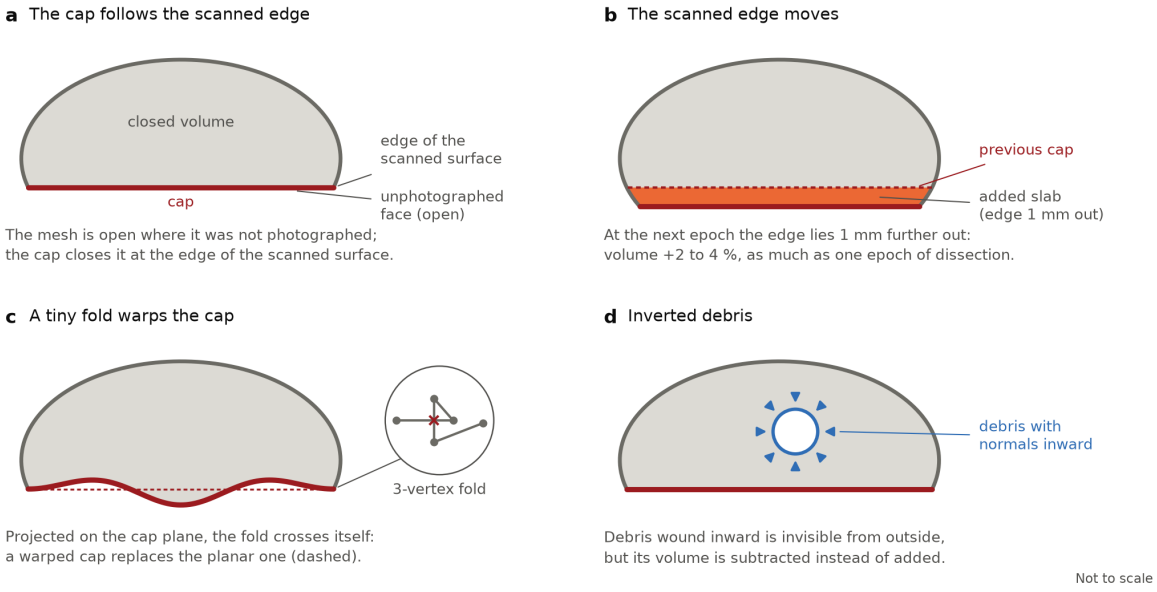


Figure 1: *

Figure 1 (`fig1_concept.png`, 420 characters). Why a watertight volume can still be wrong. Cross-section of a dissected hemisphere (grey), open on the unphotographed face. (a) The cap (red) follows the edge of the scanned surface. (b) If this edge lies 1 mm further out at the next epoch, the volume grows by 2–4%, as much as a typical epoch of dissection. (c) A three-vertex fold of the boundary warps a planar cap. (d) Inverted debris silently subtracts its volume.

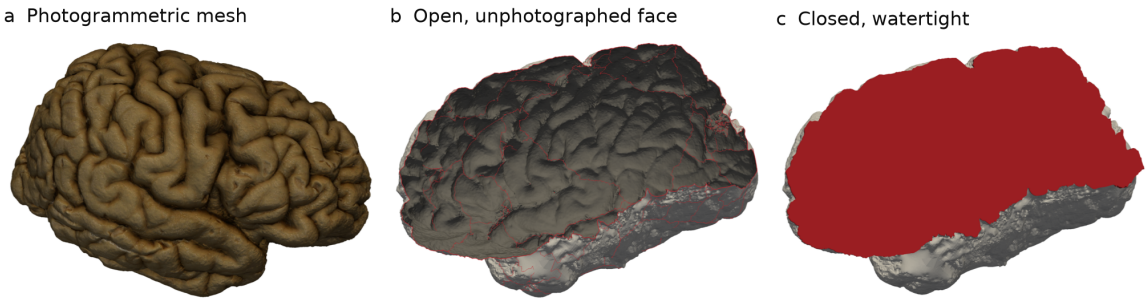


Figure 2: *

Figure 2 (`fig1_pipeline.png`, 209 characters). Pipeline on a successful epoch (spc-01, epoch 1). (a) Textured photogrammetric mesh. (b) Open boundary, seen from the unphotographed face (inside of the shell in dark grey). (c) Watertight result, caps in red.

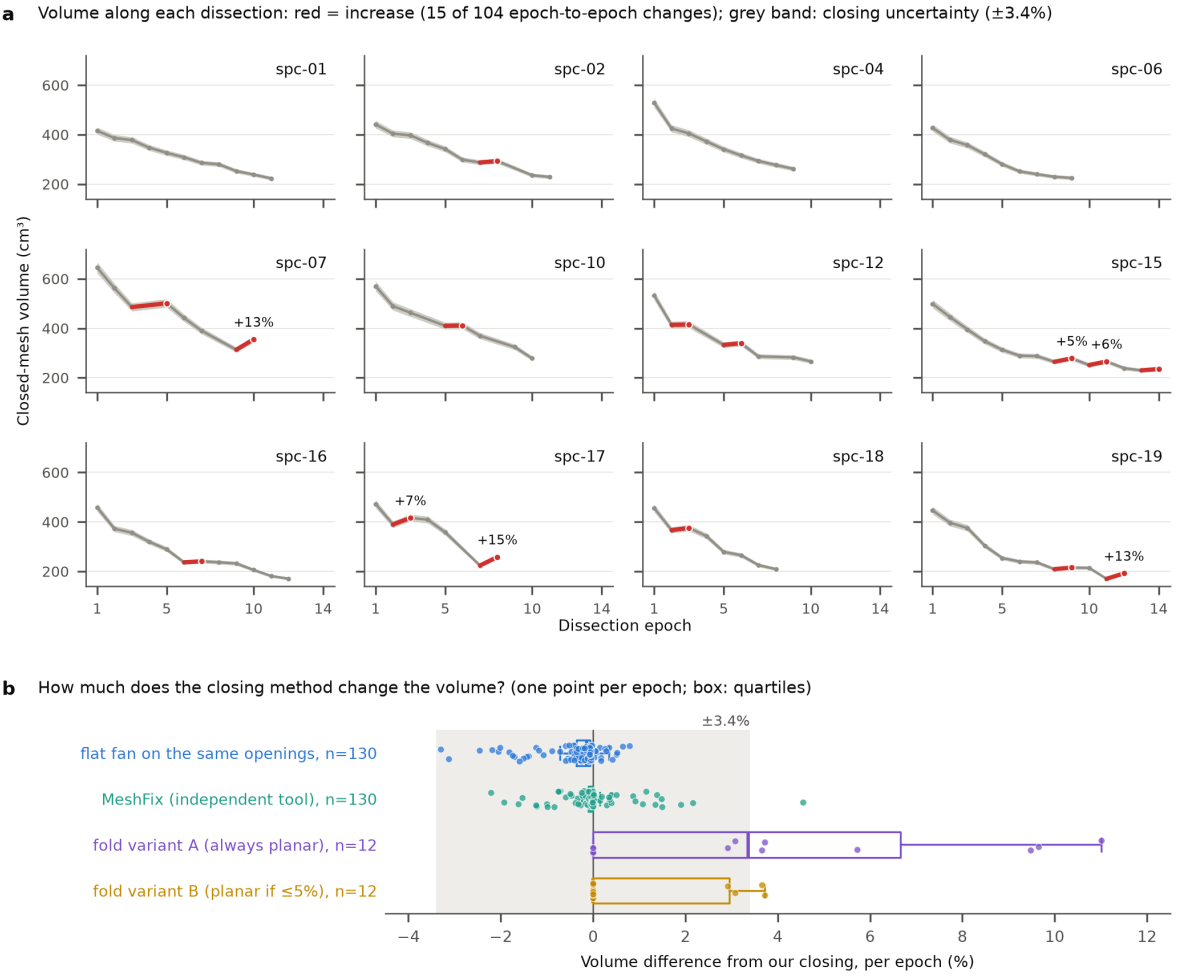


Figure 3: *

Figure 3 (`fig2_volume_vs_epoch.png`, 482 characters). (a) Closed volume versus epoch, one panel per specimen. Volume should only decrease; red: the 15 increases among the 104 epoch-to-epoch changes. Grey band: closing uncertainty, ±3.4%, the largest difference between our closing and a flat fan over all epochs; a change inside it cannot be told apart from a closing effect. (b) Volume difference between our closing and a flat fan or MeshFix (130 epochs), and the two fold-straightening variants (12 problem epochs). Boxes: quartiles.

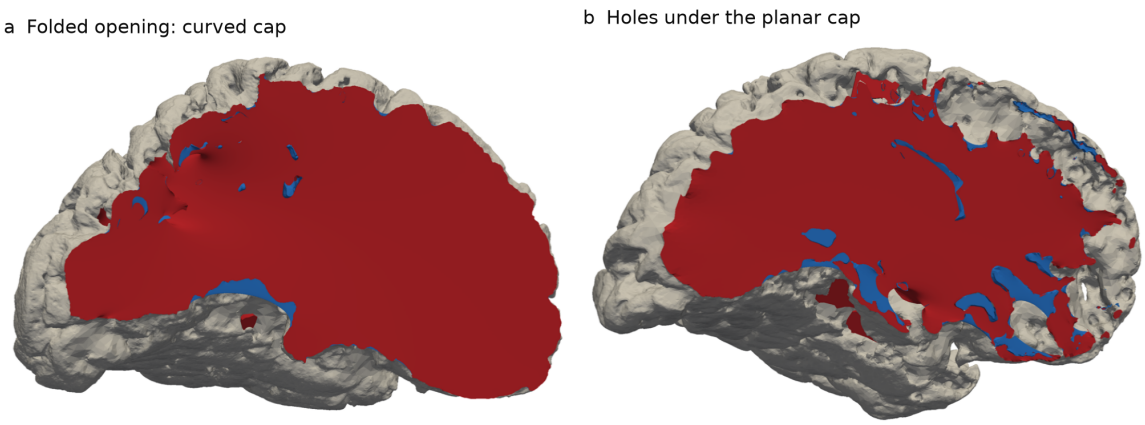


Figure 4: *

Figure 4 (`fig3_failures.png`, 183 characters). What works poorly (caps in red, back-facing surfaces in blue). (a) Folded opening (spc-07, epoch 2): the curved cap folds. (b) Small holes just under the planar cap (spc-15, epoch 8).

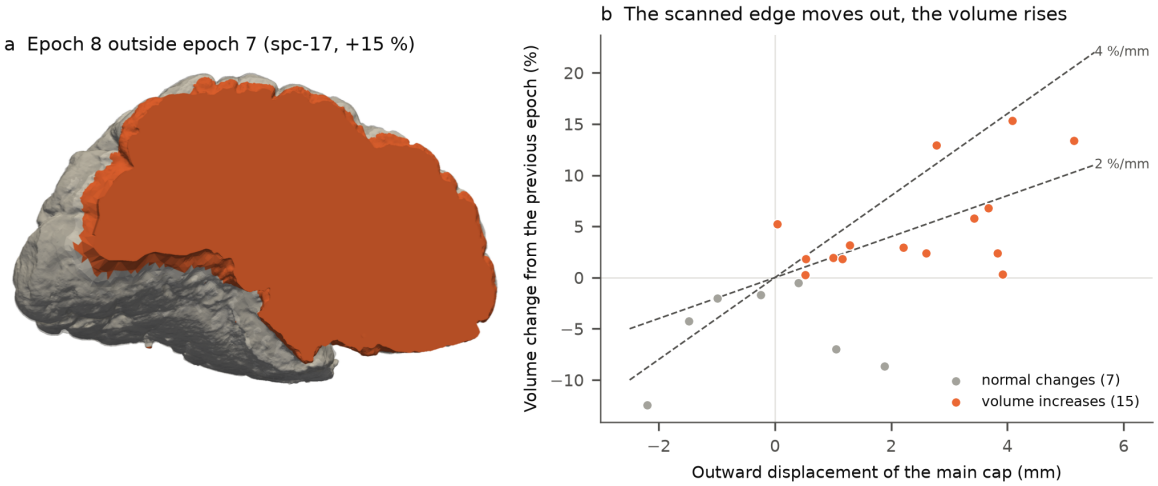
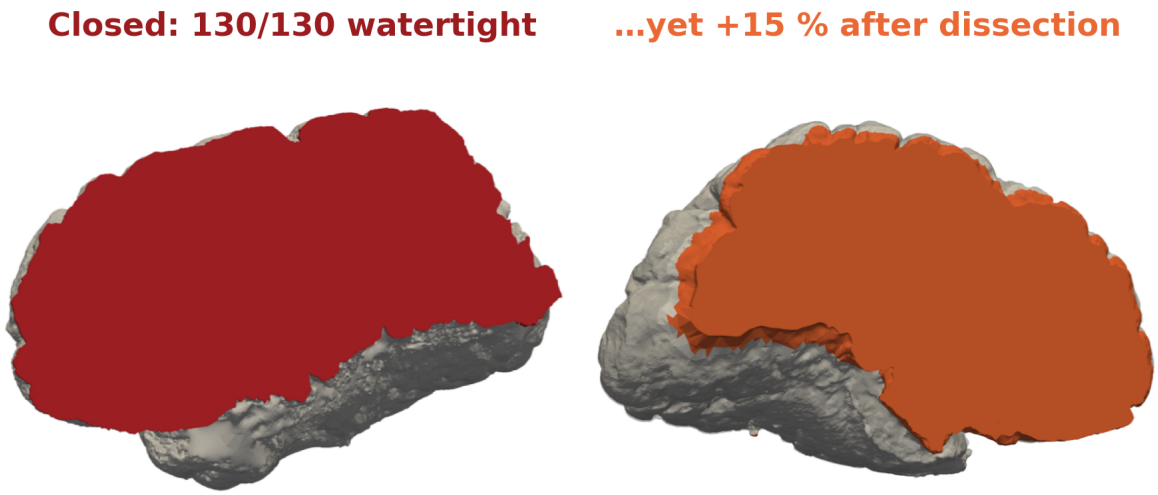


Figure 5: *

Figure 5 (`fig5_scanned_edge.png`, 443 characters). Where the volume increases come from. (a) Epoch 8 (orange) lying outside epoch 7 (grey) of spc-17: the scanned edge moved 4.1 mm outward and the volume rose by 15%. (b) Outward displacement of the main cap versus volume change, for all 15 volume increases (orange) and 7 normal decreases chosen as controls (grey). Dashed: volume added by the displacement alone (2 and 4% per mm, for cut faces of 70 to 120 cm^2); points below also lost tissue.

Preview figure



References (Vancouver, in order of first citation)

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5. Attene M. A lightweight approach to repairing digitized polygon meshes. Vis Comput. 2010;26:1393–1406. doi:10.1007/s00371-010-0416-3

Acknowledgements

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